

Linux Permissions and Security Theory (1 hour):

File and directory permissions in Linux: chmod, chown. Users, groups, and access control. Practical (1 hour): Set up user-specific access to files. Task: Demonstrate restricted access to a directory for specific users.

TASKS

1. Create two new users on your Linux system named 'musicfan' and 'moviebuff', then create a new group called 'playlist' and add both users to this group.

Create Users and Group

Step 1: Create user musicfan

```
useradd musicfan
```

Set password (Music@123)

```
passwd musicfan
```

Step 2: Create user moviebuff

```
useradd moviebuff
```

Set password (Movie@123)

```
passwd moviebuff
```

Step 3: Create group

```
groupadd playlist
```

Step 4: Add both users to playlist group

```
usermod -aG playlist musicfan
```

```
usermod -aG playlist moviebuff
```

Step 5: Verify

```
id musicfan
```

```
id moviebuff
```

```
[root@localhost ~]# id musicfan
uid=1001(musicfan) gid=1001(musicfan) groups=1001(musicfan),1003(playlist)
[root@localhost ~]# id moviebuff
uid=1002(moviebuff) gid=1002(moviebuff) groups=1002(moviebuff),1003(playlist)
```

2. Inside your home directory, create a folder called 'spotify_shared'. Set its permissions so only users in the 'playlist' group can read and write files inside it, but others cannot access it.
Hint: Use chown and chmod with the group ownership and permission bits.

Create Shared Directory

Go to root home:

```
cd /root
```

Create folder :

```
mkdir spotify_shared
```

Change group ownership:

```
chown root:playlist spotify_shared
```

Give permissions

```
chmod 770 spotify_shared
```

Verify

```
ls -ld spotify_shared
```

```
[root@localhost ~]# cd /root
[root@localhost ~]# mkdir spotify_shared
[root@localhost ~]# chown root:playlist spotify_shared
[root@localhost ~]# chmod 770 spotify_shared
[root@localhost ~]# ls -ld spotify_shared
drwxrwx---. 2 root playlist 6 Jul 14 12:42 spotify_shared
[root@localhost ~]#
```

3. Place a text file named 'top_songs.txt' inside 'spotify_shared'. Change its ownership so that 'musicfan' is the owner and 'playlist' is the group, then set permissions so only the owner can edit, group members can read, and others have no access.

Create File

Go inside directory

```
cd spotify_shared
```

Create file:

touch top_songs.txt

Open file:

vi top_songs.txt

Press: i

Write:

Shape of You

Believer

Perfect

Levitating

Blinding Lights

Press:

Esc

Save:

:wq

Change owner

chown musicfan:playlist top_songs.txt

Set permission

chmod 640 top_songs.txt

Verify

ls -l

```
[root@localhost ~]# cd spotify_shared
[root@localhost spotify_shared]# touch top_songs.txt
[root@localhost spotify_shared]# vi top_songs.txt
[root@localhost spotify_shared]# chown musicfan:playlist top_songs.txt
[root@localhost spotify_shared]# chmod 640 top_songs.txt
[root@localhost spotify_shared]# ls -l
total 4
-rw-r----- 1 musicfan playlist 57 Jul 14 12:50 top_songs.txt
[root@localhost spotify_shared]#
```

4. Simulate a scenario where 'moviebuff' tries to access 'top_songs.txt'.

Switch to the 'moviebuff' user and attempt to open and edit the file.

Describe what happens and explain why based on the permissions you set.

Step 1: Make sure you are root

Prompt:

```
[root@localhost ~]#
```

If your folder is still in /root, move it first (otherwise moviebuff cannot access /root):

```
mv /root/spotify_shared /home/spotify_shared
```

```
chown root:playlist /home/spotify_shared
```

```
chmod 770 /home/spotify_shared
```

Step 2: Switch to moviebuff

```
su - moviebuff
```

```
[root@localhost ~]# su - moviebuff
[moviebuff@localhost ~]$ cat /home/spoti
Shape of You
```

Step 3: Read the file

Run:

```
cat /home/spotify_shared/top_songs.txt
```

```
[root@localhost ~]# su - moviebuff
[moviebuff@localhost ~]$ cat /home/spotify_shared/top_songs.txt
Shape of You
Believer
Perfect
Levitating
Blinding Lights
```

Step 4: Try to edit the file

Run:

```
vi /home/spotify_shared/top_songs.txt
```

Press: i

Type: New Song

Press: Esc

Type: :wq

Press **Enter**.

```
moviebuff@localhost:~  
Shape of You  
Believer  
Perfect  
Levitating  
Blinding Lights  
New Song  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
E45: 'readonly' option is set (add ! to override)
```

Step 5: Return to root

exit

Explanation :

What happened?

- moviebuff can **read** the file because it is a member of the **playlist** group.
- moviebuff **cannot edit or save** the file because only the owner (musicfan) has write permission.
- Other users have no access.